

A formal-group proof of a conjecture of Z.-H. Sun for an Apéry-like sequence

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20 September 2026

Abstract

Let

$$G_n = \sum_{k=0}^n 4^k \binom{2n-2k}{n-k}^2 \binom{2k}{k}.$$

We prove a conjecture of Z.-H. Sun asserting a three-term congruence for $G_{(mp^r-1)/2}$ when $p \equiv 1 \pmod{4}$. The main observation is that the formal logarithm $\ell(v) = \sum_{n \geq 0} G_n v^{2n+1}/(2n+1)$ becomes, after the integral strict change of parameter $v = \sqrt{\lambda(4z)}/4$, the Eichler integral of the CM newform $\eta(4z)^6$. The Hecke recurrence of this form and the standard formal-group transfer theorem then give the conjectured congruence for every odd m , including $p \mid m$, and every $r \geq 2$.

Keywords. Apéry-like numbers; supercongruences; formal groups; CM modular forms; Atkin–Swinnerton-Dyer congruences.

2020 Mathematics Subject Classification. 11B65, 11F33, 11G20, 11A07.

1 Introduction

Z.-H. Sun introduced and studied the Apéry-like sequence

$$G_n = \sum_{k=0}^n 4^k \binom{2n-2k}{n-k}^2 \binom{2k}{k}. \quad (1)$$

Its generating series is hypergeometric, but the indices in Sun’s Conjecture 2.7 are multiplicative: one compares coefficients whose odd indices differ by powers of a prime. The bridge between these two descriptions is a one-dimensional formal group. In the correct integral coordinate, its logarithm is the Eichler integral of a CM eigenform, so the three-term Hecke recurrence becomes the desired coefficient congruence.

We prove the following statement conjectured in [2].

Theorem 1.1. *If $p \equiv 1 \pmod{4}$ is prime, write $p = x^2 + 4y^2$. Then, for every positive odd integer m and every $r \geq 2$,*

$$G_{(mp^r-1)/2} \equiv (4x^2 - 2p)G_{(mp^{r-1}-1)/2} - p^2 G_{(mp^{r-2}-1)/2} \pmod{p^r}. \quad (2)$$

The coefficient $4x^2 - 2p$ is therefore not an isolated numerical pattern: it is the p th Hecke eigenvalue of the weight-three CM newform $\eta(4z)^6$. The proof has three steps. First, the defining convolution gives an elementary hypergeometric generating function. Second, the strict integral parameter $v = \sqrt{\lambda(4z)}/4$ identifies the associated formal logarithm with the Eichler integral of that newform. Finally, the formal-group transfer theorem carries its

Hecke recurrence back to the coefficients G_n , with exactly the modulus p^r . Sections 2–4 follow these three steps in order.

The argument also gives, for $p \equiv 3 \pmod{4}$, the baseline congruence

$$G_{(mp^r-1)/2} \equiv p^2 G_{(mp^{r-2}-1)/2} \pmod{p^r}.$$

This is weaker than the modulus p^{2r-1} in Sun's Conjecture 2.6 and is recorded only to delimit the result proved here.

2 The generating function and its modular parameter

Put $F(t) = \sum_{n \geq 0} G_n t^n$. We first give the elementary generating function identity needed later.

Lemma 2.1. *One has*

$$F(t) = \frac{{}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; 16t)}{\sqrt{1-16t}}. \quad (3)$$

Consequently, with the usual modular lambda function,

$$F\left(\frac{\lambda(\tau)}{16}\right) = \frac{\theta_3(\tau)^4}{\theta_4(\tau)^2}. \quad (4)$$

Proof. The convolution in (1) gives

$$F(t) = \left(\sum_{j \geq 0} \binom{2j}{j}^2 t^j \right) \left(\sum_{k \geq 0} \binom{2k}{k} (4t)^k \right).$$

The two factors are respectively ${}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; 16t)$ and $(1-16t)^{-1/2}$, which proves (3). Now use the classical identities

$${}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; \lambda(\tau)\right) = \theta_3(\tau)^2, \quad 1 - \lambda(\tau) = \frac{\theta_4(\tau)^4}{\theta_3(\tau)^4}.$$

□

Define the odd formal series

$$\ell(v) = \sum_{n \geq 0} \frac{G_n}{2n+1} v^{2n+1}; \quad \ell'(v) = F(v^2). \quad (5)$$

Write $q = e^{2\pi iz}$ and set

$$v(q) = \frac{\sqrt{\lambda(4z)}}{4} = q \prod_{n \geq 1} \left(\frac{1+q^{4n}}{1+q^{4n-2}} \right)^4. \quad (6)$$

The product shows that $v(q) \in q + q^3 \mathbb{Z}[[q^2]]$; hence it is a strict integral local parameter and its compositional inverse also belongs to $v + v^2 \mathbb{Z}[[v]]$.

Proposition 2.2. *If $\eta(4z)^6 = \sum_{n \geq 1} a_n q^n$, then*

$$\ell(v(q)) = \sum_{n \geq 1} \frac{a_n}{n} q^n. \quad (7)$$

Proof. The standard differential identity

$$q \frac{d}{dq} \lambda(4z) = 2\theta_3(4z)^4 \lambda(4z)(1 - \lambda(4z))$$

and (4) give

$$q \frac{d}{dq} \ell(v(q)) = \ell'(v(q)) q \frac{dv(q)}{dq} = \frac{\theta_3(4z)^4}{\theta_4(4z)^2} v(q) \theta_4(4z)^4.$$

Since $v(q) = \theta_2(4z)^2 / (4\theta_3(4z)^2)$ and $\theta_2\theta_3\theta_4 = 2\eta^3$, the last expression is $\eta(4z)^6$. Both sides of (7) have zero constant term, so termwise integration proves the assertion. $\square$

3 The CM newform

The eta product

$$f(z) = \eta(4z)^6 = \sum_{n \geq 1} a_n q^n$$

is the normalized newform of weight 3, level 16, and character χ_{-4} . It has complex multiplication by $\mathbb{Q}(i)$. Equivalently it is the theta series of the Hecke character of $\mathbb{Q}(i)$ of infinity type 2 and conductor (2), with normalization $a_1 = 1$. It follows that, for every odd prime p ,

$$a_{pn} - a_p a_n + \chi_{-4}(p) p^2 a_{n/p} = 0, \quad (8)$$

where $a_{n/p} = 0$ when $p \nmid n$, and

$$a_p = \begin{cases} 4x^2 - 2p, & p = x^2 + 4y^2 \equiv 1 \pmod{4}, \\ 0, & p \equiv 3 \pmod{4}. \end{cases} \quad (9)$$

For the split case, if $\pi = x + 2iy$ is chosen primary, the Hecke value is $\pi^2 + \bar{\pi}^2 = 2(x^2 - 4y^2) = 4x^2 - 2p$.

4 Formal-group transfer

We recall the coefficient form of the standard transfer theorem; see Li and Long [1, Proposition 3 and Theorem 6].

Theorem 4.1 (formal-group coefficient transfer). *Let p be odd and let*

$$L(q) = \sum_{n \geq 1} \frac{a_n}{n} q^n, \quad \ell(v) = \sum_{n \geq 1} \frac{b_n}{n} v^n$$

be related by $\ell(v(q)) = L(q)$, where $v(q) \in q + q^2 \mathbb{Z}_{(p)}[[q]]$. Suppose

$$L(q) - \frac{A}{p} L(q^p) + B L(q^{p^2}) \in \mathbb{Z}_{(p)}[[q]]. \quad (10)$$

Then, for all positive integers m and $r \geq 2$,

$$b_{mp^r} - A b_{mp^{r-1}} + p^2 B b_{mp^{r-2}} \equiv 0 \pmod{p^r}. \quad (11)$$

No coprimality condition on m is required.

For clarity, (10) is the functional-equation-lemma description of a one-dimensional formal group over $\mathbb{Z}_{(p)}$. Its Cartier parameters are $\mu_0 = A$, $\mu_1 = -pB$, and $\mu_j = 0$ for $j \geq 2$. A strict integral change of coordinate preserves the formal group, and the coefficient congruence in (11) is precisely the resulting Cartier congruence. This also explains why the modulus is p^r and why $p \mid m$ is allowed.

Proof of the main result. Take $L(q) = \sum a_n q^n / n$. Relation (8) implies

$$L(q) - \frac{a_p}{p} L(q^p) + \chi_{-4}(p) L(q^{p^2}) \in \mathbb{Z}_{(p)}[[q]]. \quad (12)$$

Indeed, comparison at an exponent divisible by p is exactly (8), while the remaining coefficients are p -integral. By Proposition 2.2, Theorem 4.1 applies through the strict parameter (6) with $A = a_p$ and $B = \chi_{-4}(p)$.

In (5), the coefficient b_N is zero for even N and equals $G_{(N-1)/2}$ for odd N . If $p \equiv 1 \pmod{4}$, then $\chi_{-4}(p) = 1$ and (9) gives $a_p = 4x^2 - 2p$. Substitution in (11), for odd m , is exactly (2). $\square$

Remark 4.2. For $p \equiv 3 \pmod{4}$, the same proof gives

$$G_{(mp^r-1)/2} \equiv p^2 G_{(mp^{r-2}-1)/2} \pmod{p^r},$$

because $a_p = 0$ and $\chi_{-4}(p) = -1$. The sign is positive after moving the last term in (11) to the other side.

Computational verification

Exact rational power-series calculations were used only as diagnostics. The accompanying program verifies through q^{99} that the parameter (6) is integral and odd and that $q d\ell(v(q))/dq = \eta(4z)^6$. It also checks (9) for every odd prime below 100. None of these finite checks replaces a step of the proof.

Use of generative artificial intelligence

The proof presented in this article was found by OpenAI Codex.

Declarations

Funding. No funding was received for this work.

Competing interests. The author declares no competing interests.

Data availability. The exact verification program is available at the public GitHub repository.

References

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